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Systems Design

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Summer

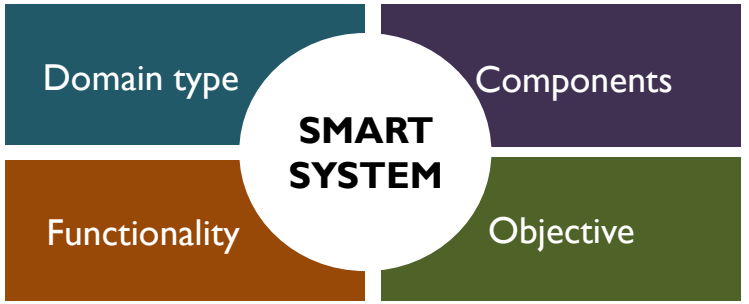
SYSTEM DEVELOPMENT PROCESS

Phases

- System introduction
- System analysis
- System architecture design
- System implementation

System introduction

- Project scope
  - Background information
  - Problem statement
- Project goals



System analysis

- What do the users need and want from the system?
- Deliverable is a list of requirements that will fulfil the system goals
- Two tasks
  - Identify and express system requirements
  - Prioritise system requirements

Identify and express system requirements

- System users are primary source of requirements

- Types of requirements
  - Functional requirements
  - Non-functional requirements
  - Domain-oriented requirements [1]
  - User-related requirements [2]

- Ways of expression
  - Use cases (here)
  - User stories (here)

[1] Georgievski, I. (2023) Software Development Lifecycle for Engineering AI Planning Systems. In International Conference on Software Technologies, pages 751–760.
[2] Aiello, M., Fiorini, L., and Georgievski, I. (2021) Software Engineering Smart Energy Systems. Handbook of Smart Energy Systems, pages 1–21.

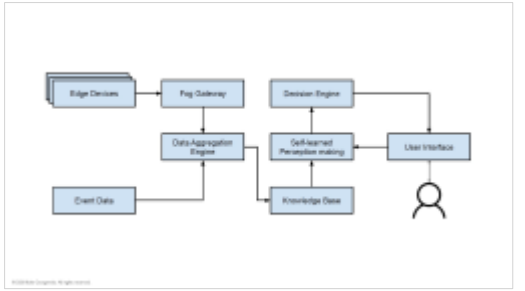
Prioritise system requirements

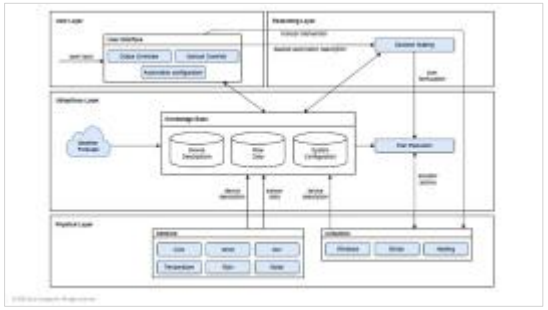
- Mandatory requirements
- Desirable requirements

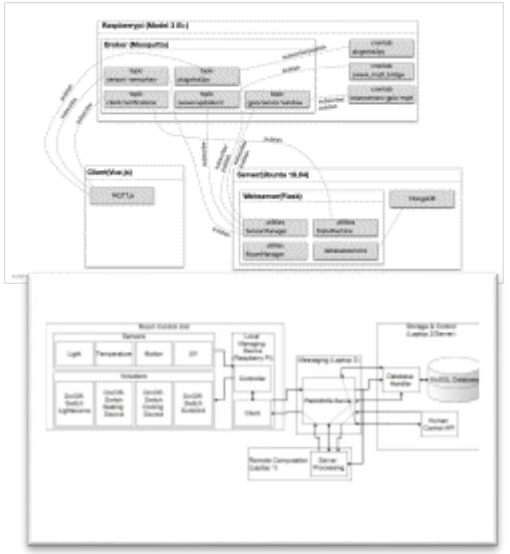
System architecture design

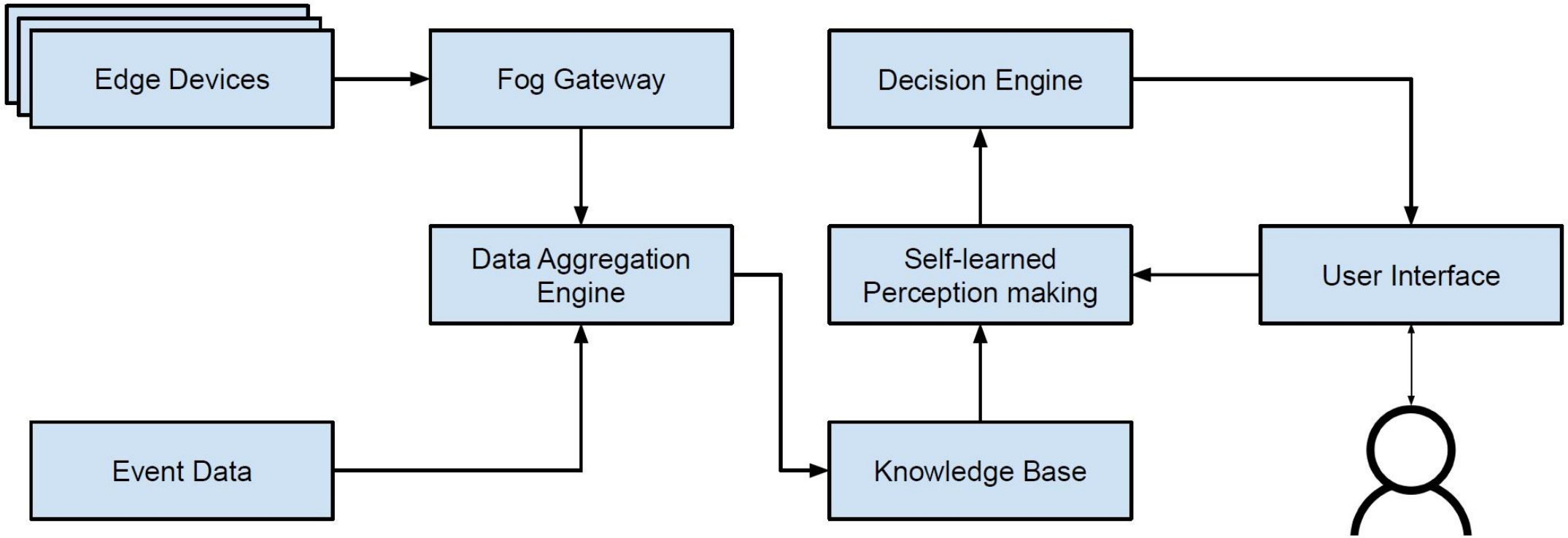
- What a system is or does
- Not how a system is technically implemented
- Implementation independent

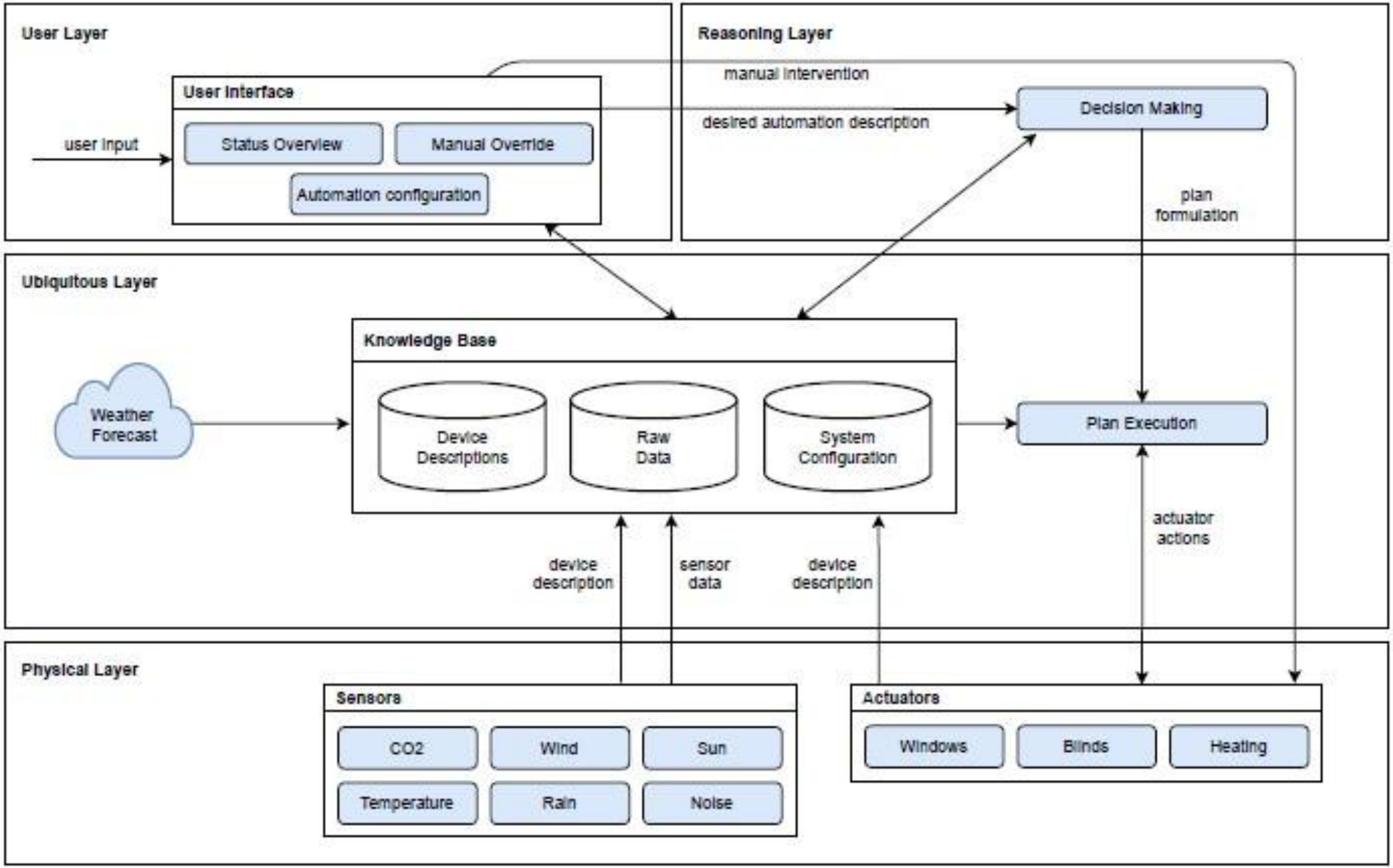
- Tasks
  - Identify system components
  - Connect system components and put them in layers
  - Model the system architecture design

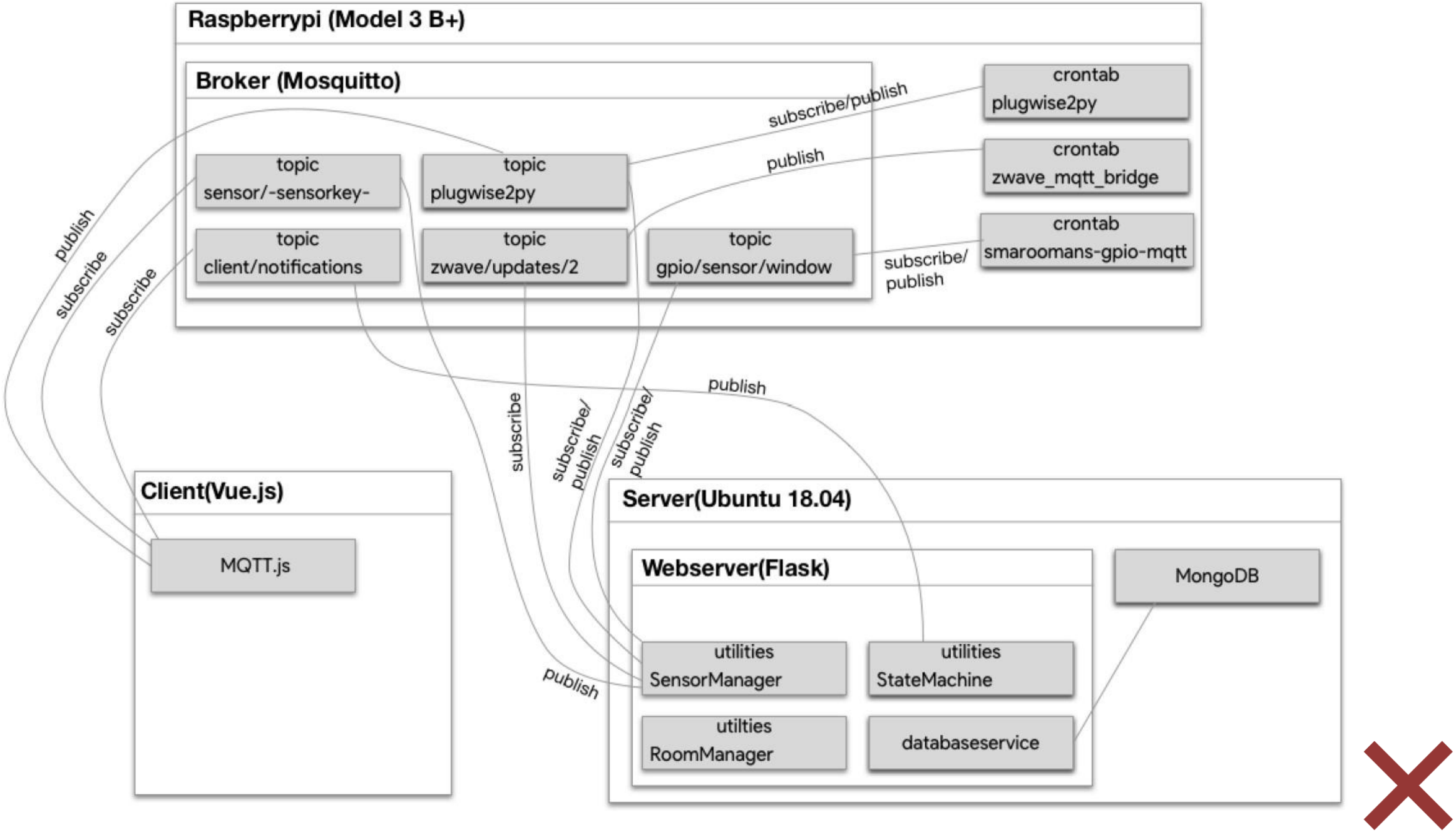


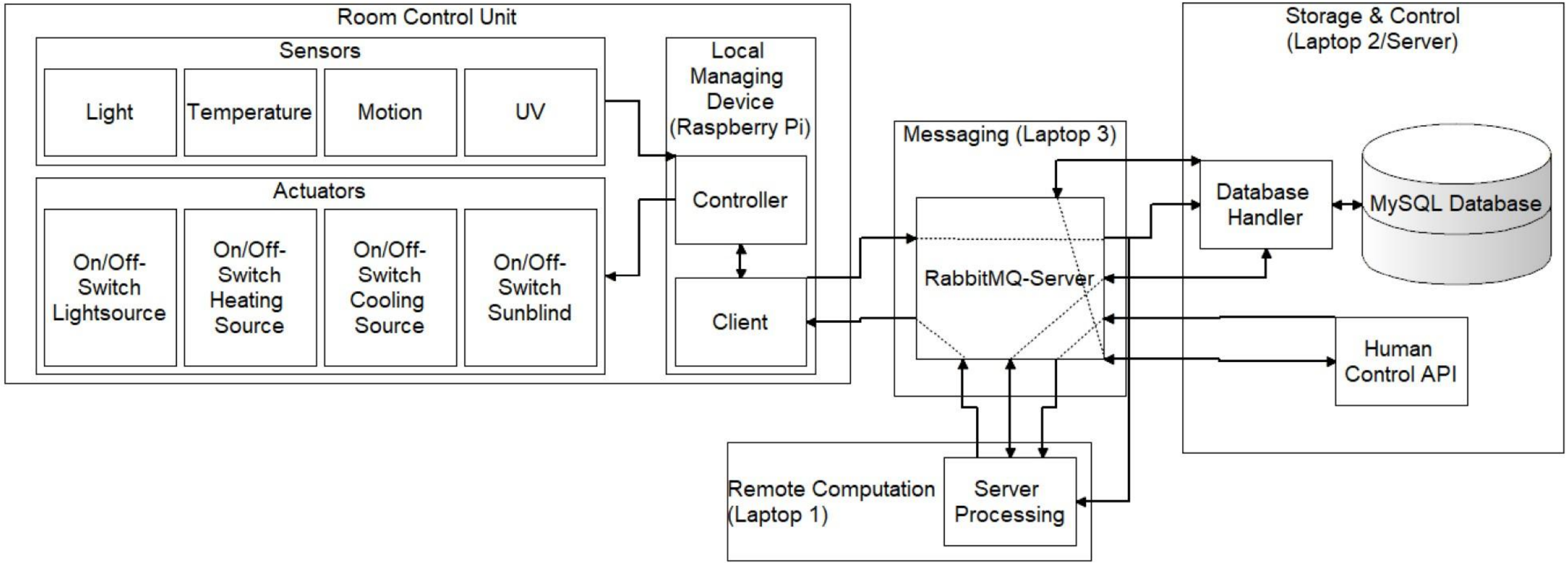





Edge Devices
Fog Gateway
Decision Engine
Data Aggregation Engine
Self-learned Perception making
User Interface
Event Data
Knowledge Base


User Layer
User Interface
Status Overview
Manual Override
Automation configuration
user input
Reasoning Layer
manual intervention
desired automation description
Decision Making
plan formulation
Ubiquitous Layer
Knowledge Base
Device Descriptions
Raw Data
System Configuration
Weather Forecast
Plan Execution
actuator actions
device description
sensor data
device description
Physical Layer
Sensors
CO2
Wind
Sun
Temperature
Rain
Noise
Actuators
Windows
Blinds
Heating


Raspberrypi (Model 3 B+)
Broker (Mosquitto)
topic
sensor/-sensorkey-
topic
plugwise2py
topic
client/notifications
topic
zwave/updates/2
topic
gpio/sensor/window
crontab
plugwise2py
crontab
zwave_mqtt_bridge
crontab
smaroomans-gpio-mqtt
subscribe/publish
publish
subscribe/
publish
publish
subscribe
subscribe
publish
subscribe
subscribe/
publish
subscribe/
publish
publish
Client(Vue.js)
MQTT.js
Server(Ubuntu 18.04)
Webserver(Flask)
utilities
SensorManager
utilities
StateMachine
utilties
RoomManager
databaseservice
MongoDB


Room Control Unit
Sensors
Light
Temperature
Motion
UV
Actuators
On/Off-Switch Lightsource
On/Off-Switch Heating Source
On/Off-Switch Cooling Source
On/Off-Switch Sunblind
Local Managing Device (Raspberry Pi)
Controller
Client
Messaging (Laptop 3)
RabbitMQ-Server
Storage & Control (Laptop 2/Server)
Database Handler
MySQL Database
Human Control API
Remote Computation (Laptop 1)
Server Processing

Identify components

- Component
  - Specific capability/functionality
  - Specific task
  - Group of related functions
- Components loosely coupled
- Components functionally cohesive

Separation of Concerns (SoC)

- Organising a system into distinct components
  - Each with a specific concern

- Important in smart systems
  - Modularity
  - Scalability and flexibility
  - Improved collaborations
  - Simplifies complexity
  - Better fault tolerance
  - …

Connect and layer

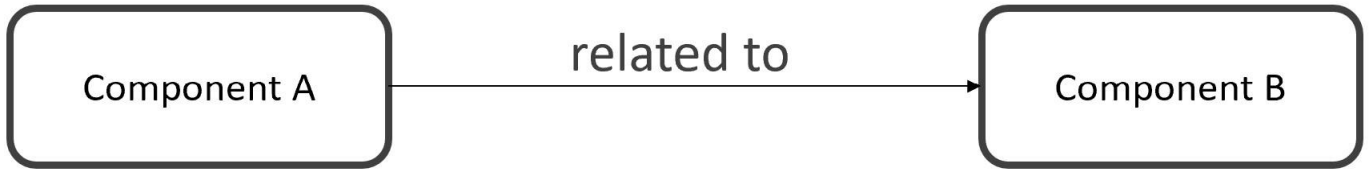
- Presentation layer
- Reasoning layer
- Ubiquitous layer
- Physical layer

Layers of a typical application:

- Presentation layer
- Application logic layer
- Data layer

Modelling

- Data flow diagram
  - Tutorial here
- UML Component diagram
  - Tutorial here
- Simple box-and-line diagram



System implementation

- Implement each system component
- Construct the whole system and put it into operation
- Test
  - System components individually tested
  - Complete system tested
- Tools and methods in subsequent practicals

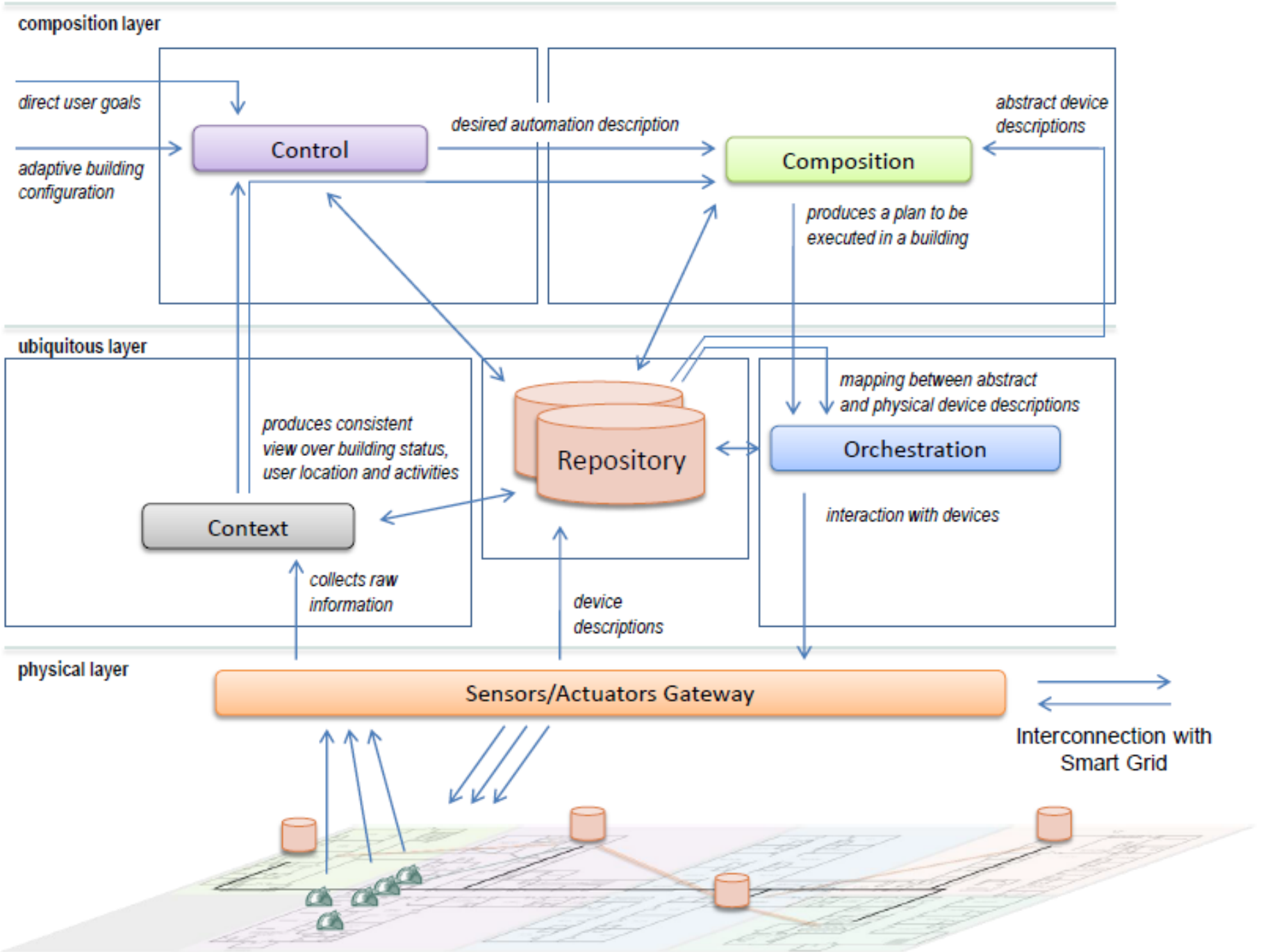
Project proposal/Report

- Project information
- System focus
- Project requirements
- System architecture design

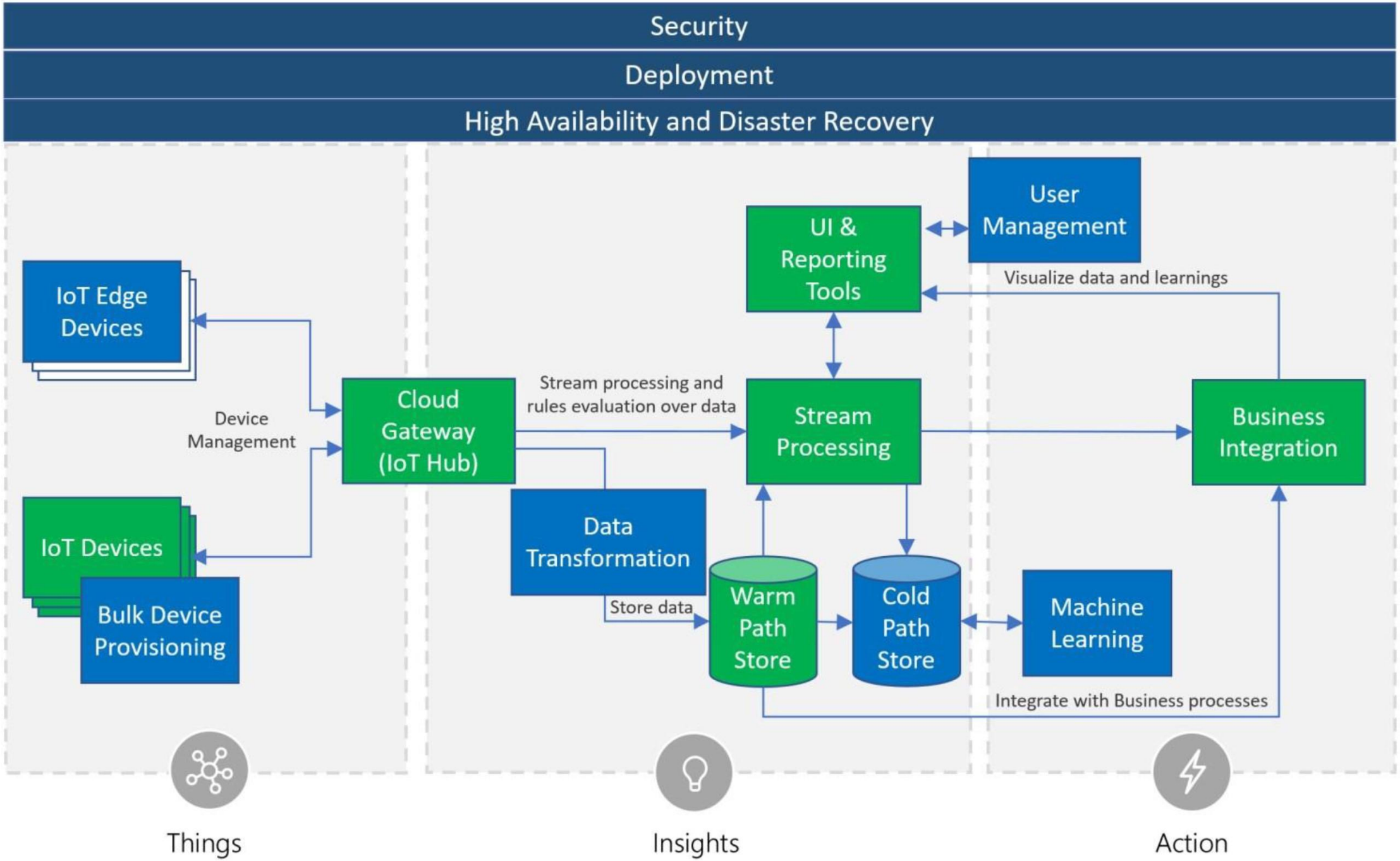
- System implementation

– Give a link to code repository (e.g., GitHub, Bitbucket, etc.)

EXAMPLES OF SYSTEM ARCHITECTURE DESIGNS FOR SMART SYSTEMS

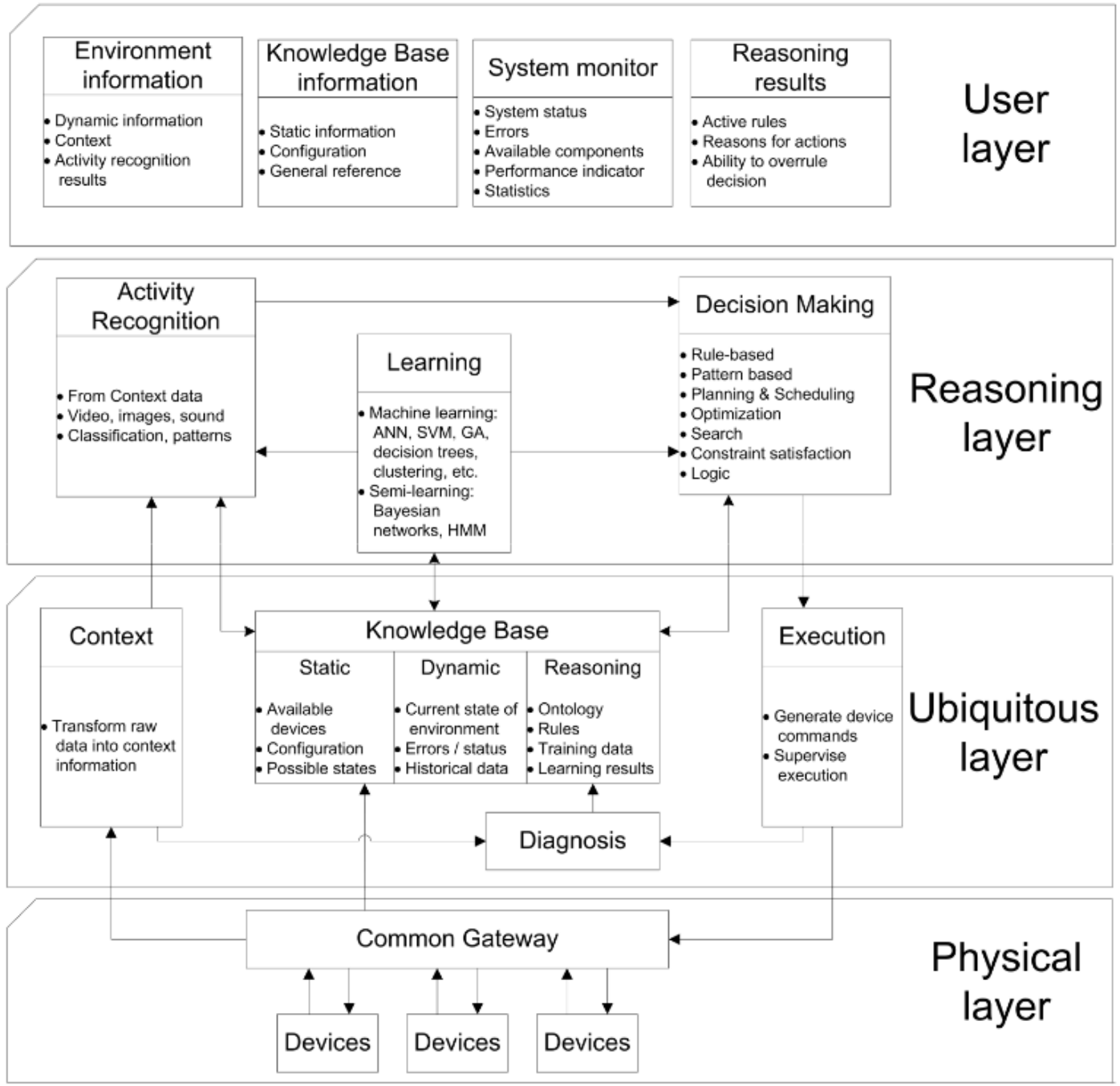


Degeler, V., Gonzalez, L. I. L., Leva, M., Shrubsole, P., Bonomi, S., Amft, O., & Lazovik, A. (2013) Service-oriented architecture for smart environments. In IEEE International Conference on Service-Oriented Computing and Applications, pp. 99-104.



Microsoft. (2018) *Microsoft Azure IoT Reference Architecture*. Version 2.1.

Figure 1: Smart System Architecture Pattern



Degeler, V. and Lazovik, A. (2013) *Architecture pattern for context-aware smart environments*. In Creating Personal, Social and Urban Awareness through Pervasive Computing. IGI Global.

SYSTEM DESIGN DECISIONS

Examples of *what* questions

- What kind of non-residential buildings/autonomous ground vehicles is the system for?
- What types of data describe the system?
- What types of data models characterise the system?
- What kinds of metadata are needed for the system?
- What types of data need to be stored and accessed by the system?
- What context describe the system?
- What kinds of user needs are included in the system?
- What types of control are included in the system?
- What kind of interaction the user can have with the system?
- What information will the system show to user?
- What will an edge component do?
- What are the essential capabilities of the system?

What types of data describe the system?

- Environmental conditions
  - Temperature
  - Humidity
  - CO_2 level
  - …
- Device status
  - Light on
  - Radiator valve on
  - Window blinds down
  - …
- …

What types of data models characterise the system?

- Telemetry
- Metadata

What kinds of metadata are needed for the system?

- Device metadata
- Building metadata

What types of data need to be stored and accessed?

- Dynamic data
  - Raw stream of incoming data from a device
  - Specific values extracted from the raw data
  - Processed data
- Static data

What kinds of user needs are included in the system?

- Comfortable working conditions per type of user activity
- Healthy working habits
- Save energy
- …

What information will the system show to user?

- Live device data
- Analysis results
- Command and control capabilities
- Notifications
- …

Examples of *how* questions

- How is the data collected from IoT formatted?
- How are messages specified and stored?
- How is the metadata specified?
- How is static data stored and accessed?
- How will components communicate among each other?
- How to implement automation?

How is the metadata specified?

- Device metadata
  - Identifier
  - Class or type
  - Model
  - Hardware serial number
  - …
- Building metadata
  - Floors
  - Rooms
  - Types of rooms
  - Device location
  - …

How is the data collected from IoT formatted?

- Message
  - Type ID
  - Instance ID
  - Timestamp
  - Value

How to implement automation?

- AI planning
  - Planning domain model
  - Planning problem instance
  - Planner

Icons designed by Freepik.